CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of MSc Microbiology Examination April/May 2019

Course code & Name MI 761 & Microbial Genetics

Date: 4/5/2019 Day: Saturday Time: 10:00 am To 10:30 am Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.		20
1.	________ is the sub-unit of Cas9, which binds with the guide RNA.		
	(i) Rec II	(ii) PMP interacting domain	
	(iii) Rec I	(iv) RuvC	
2.	Q acts as an anti-terminator for transcription of ____ genes during lytic cycle.		
	(i) Late	(ii) Immediate Early	
	(iii) Delayed Early	(iv) All of the above	
3.	_____ never leads to structural abnormality.		
	(i) Deletion	(ii) Inversion	
	(iii) Translocation	(iv) Nondisjunction	
4.	The unit for gene mapping during interrupted mating experiment is _____		
	(i) Bp	(ii) kD	
	(iii) Time	(iv) All of the above	
5.	________ is the longest transposable element known.		

	(i) Non composite transposons	(ii) Composite transposons	
	(iii) Bacteriophage μ	(iv) IS elements	
6.	**_____ is true about LINES.**		
	(i) LTR Interspersed repeat sequence	(ii) Non-LTR Interspersed repeat sequence	
	(iii) DNA transposons	(iv) Satellite DNA	
7.	**During conjugation, OmpA of the recipient cell interacts with ____ of the donor cell to establish stable mating.**		
	(i) Tra D	(ii) Tra N	
	(iii) Tra G	(iv) Both, (ii) and (iii)	
8.	**Conjugation between F^+ and F^- cells always leads to generation _____ cells.**		
	(i) F^+	(ii) F^-	
	(iii) Hfr	(iv)F’	
9.	**Purine is replaced by Purine due to mutation. It is called as _____.**		
	(i) Transition	(ii) Transversion	
	(iii) Transposition	(iv) Transfection	
10.	**During deamination, Adenine converts to ______**		
	(i) Thymine	(ii) Uracil	
	(iii) Xanthine	(iv) Hypoxanthine	
11.	**_________ level of gene regulation is not evident in *E. coli*.**		
	(i) Transcription	(ii) RNA processing	
	(iii) Chromatin remodeling	(iv) None of the above	
12.	**__________ codes for monocystronic m-RNA.**		
	(i) E. coli	*(ii) S. cerevisiae*	
	(iii) B. subtilis	(iv) Both, (i) and (iii)	
13.	**Which of the following is the most appropriate definition of an operator?**		
	(i) A non-coding, regulatory DNA sequence that is bound by RNA polymerase	(ii) A non-coding, regulatory DNA sequence that is bound by a repressor protein	
	(iii) A DNA-binding protein that regulates gene expression	(iv) A cluster of genes that are regulated by a single promoter	
14.	**Transfer of a gene in animal cell is called as ____.**		
	(i) Transformation	(ii) Conjugation	
	(iii) Both, (i) and (ii)	(iv) Transfection	

15.	**____ resolves the holiday junction model.**		
	(i) Ruv A	(ii) Ruv B	
	(iii) Ruv C	(iv) All of the above	
16.	**Which segments of the attenuator together form a loop during reduced concentration of tryptophan?**		
	(i) Segment 1-2	(ii) Segment 2-3	
	(iii) Segment 3-4	(iv) All of the above	
17.	**_____ is not a structural gene in the Lac operon.**		
	(i) Lac I	(ii) Lac Z	
	(iii) Lac Y	(iv) Lac A	
18.	**_________ introduces gene directly in the nucleus of an animal cell.**		
	(i) Microinjection	(ii) Protoplast fusion	
	(iii) Cell squeezing	(iv) Use of dendrimers	
19.	**____ secretes virulence factor during natural transformation.**		
	(i) Gram negative bacteria	(ii) Gram positive bacteria	
	(iii) Both, (i) and (ii)	(iv)None of the above	
20.	**During nucleotide excision repair, _____ acts as helicase.**		
	(i) uvrA	(ii) uvrB	
	(iii) uvrC	(iv) uvrD	

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II Semester of MSc Microbiology Examination April/May 2019

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Date: 4/5/2019 Day: Saturday Time: 10:31 am To 1:00 pm Maximum Marks: 50

Instructions:

1. **Section I and II must be attempted in TWO ANSWER SHEETS.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION –I		
Q - II	**Answer the following questions as directed (Any 10)**	**20**
1	Briefly explain base analog as a potent mutagen (with a suitable example).	**02**
2	What is parasexual cycle in fungi?	**02**
3	Define: Mutation rate and Mutant frequency	**02**
4	Define: Tetrad in reference to mycology	**02**
5	Differentiate: complete and abortive transduction?	**02**
6	Define: conservative, replicative and retrotransposons	**02**
7	What is immunity to superinfection during bacteriophage infection?	**02**
8	Explain natural transformation in *Haemophilus influenzae*.	**02**
9	The Hfr strain possessing the markers his^+ met^+ tyr^+ str^s was mated with a F^- strain possessing the markers his^- met^- tyr^- str^r. The time of entry for each marker is shown below. Donor Marker — Time (min) his^+ — 21 met^+ — 12 tyr^+ — 46 Based on the above results, • Draw a map of the Hfr chromosome. • Indicate the position of the F element, and the first marker to be transferred in the same.	**02**
10	What is photoreactivation?	**02**
11	What is the difference between homologous and site specific recombination?	**02**
12	Define: Latent period, Eclipse period and burst size.	**02**

SECTION – II		
Q-III	**Answer the following questions as directed (Any 6)**	**30**
1	Elucidate the regulatory mechanism involved during conjugation between F^+ and F^- cells.	**05**
2	What is natural transformation? How do streptococci uptake foreign DNA naturally?	**05**
3	Explain virulence operon in the Ti-Plasmid.	**05**
4	Write a detailed note on regulation of Lac operon, when… (a) Lactose is absent (b) Lactose is present as sole carbon source (c) Both, Lactose and Glucose are present in the media	**05**
5	Write a detailed note on various methods used in transfection.	**05**
6	Write a note on phage therapy.	**05**
7	Elucidate on the CRISPR-Cas9 technology.	**05**